

*Reclassification footnote (2026-05-26 EOD addendum, binding 2026-05-26)* This paper was drafted prior to the 2026-05-26 C1/C2/C3 epistemic-category reclassification (Math411-AddB §10 operator binding). Post-reclassification, the canonical TECT positioning is: *TECT is an emergent vacuum-and-gravity framework (7 C1 pillars at T6/T7) with an unresolved gauge-sector completion (Pillar 6 standard routes EXHAUSTED) and phenomenological dark-matter/quantum-origin pillars open.* The TOE-level claim is SUSPENDED until a non-standard rescue is independently derived. Pillar 6 closure programme: all four standard promotion routes (Pathway B SO(4) cubic-irrelevance,  $T_{2u}$  doubling, Stueckelberg-direct, Pathway A direct SO(10)) are REFUTED or STRUCTURALLY INSUFFICIENT (Math410, Math410-AddA, Math411, Math411-AddA, Math411-AddB). Pillar 11.B is DOUBLY BLOCKED at TECT-natural (Math412-AddA, Math412-AddB). Any wording in this paper stronger than its present tier supports per Docs/status/TOE-FACT-SHEET.md should be re-calibrated against Math411-AddB §10.

## **Flat-Cartan Forcing Audit Cluster: Methodological Refinement via Multiple Audit-Rollbacks**

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During 2026-04-28 (research Rounds 9-14, the flat-Cartan forcing audit cluster), the TECT programme discovers, documents, and corrects several over-claims in the Pillar 4 realisation-forcing programme. The initial claim (Math203) that "canonical Cartan-only choice forces  $c_1 = 0$  uniquely" is FALSIFIED by Math209 explicit-transition verification (Type B equivariance violation invalidates the forcing argument). Math200 (RGE scale-coherence) is AUDIT-FLAGGED due to convention-dependent Higgs-mass offset ambiguity; Math200-AddA/B/C correct the RGE scope and establish functional  $\gamma_h(\mu)$ . Math204 (SU(5) Cartan-only forcing) is AUDIT-FLAGGED as premature (canonical choice is not yet provably forcing, contra initial claim). Math205 synthesis is undermined by Math203 FALSIFIED verdict. However, Math202 ( $H^1(\mathbb{CP}^2, \mathcal{O}) = 0 + \text{Pic classification}$ ) and Math207 (RR1 chirality-flip rigor closure) are retained as PROVED UNCONDITIONAL results. The cluster culminates in Math209 (Type B verdict: RR1b OPEN, RR1a CLOSED) and explicit-transition verification demonstrating that the three-patch Čech construction (Math162-167) holds under equivariant scrutiny, validating the Pillar 4 atlas structure despite the canonical-realisation forcing remaining underdetermined. This Epoch exemplifies honest audit discipline: over-claims are caught, corrected, and documented; the result is higher confidence in the surviving theorems and a clear roadmap for future work (Epochs 13+, refined forcing via multi-shell

RGE + BCC-defect-lattice analysis).

## I. INTRODUCTION

The Pillar 4 programme has undergone three levels of refinement: sub-task 1 (atlas closure, Epochs 9-10, T6 PROVED CONDITIONAL), sub-task 2 (canonical realisation forcing via RGE, Epoch 11 R7-R8, T6 PROVED CONDITIONAL with documented caveat). Epoch 12 applies the sharpest scrutiny: the six-round flat-Cartan forcing audit cluster (Rounds 9-14) re-examines every assumption in the realisation-forcing logic, with the explicit goal of identifying and correcting any hidden over-claims before the Stage-2 synthesis is promoted.

The result is a six-count audit-rollback event (Math200, Math203, Math204, Math205 AUDIT-FLAGGED or FALSIFIED; Math202, Math207 retained), demonstrating that TECT's audit discipline is working as designed: incorrect arguments are caught, corrected, and replaced with honest alternatives. The three-patch Čech atlas (Math162-167) survives intact, validating the geometric structure. The canonical-realisation choice remains underdetermined but is now honestly documented as such.

## II. THE FLAT-CARTAN PROGRAMME AND MOTIVATION

The flat-Cartan forcing programme asks: given the three-patch Čech atlas (Math162-167), is the choice of principal connection (via the Cartan subgroup  $H = U(1)_\chi$  inside  $SO(10)$ ) uniquely forced by topological or gauge-theoretic consistency? If forced, the canonical realisation becomes “rigid” and requires no further assumption. If not forced, the framework admits multiple equally valid realisations, and the choice must be made on other grounds (e.g., minimal action, RGE stability, etc.).

Initial investigations (Math203, Math204) claim that the forcing is complete (rigid realisation). Later audit (Math209) discovers that the claim is too strong: the Type B equivariance constraint is violated at certain transition points, opening a closure gap. The honest response is to downgrade the claim, document the gap, and continue research.

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### III. THE AUDIT ROLLBACK SEQUENCE

*a. Math200 (RGE Scale-Coherence): AUDIT-FLAGGED* Math200 attempts to enforce Cartan-only choice by requiring that the RGE flow preserves the flat-connection structure across scale-running. The result depends critically on the convention for Higgs-mass offset in the RGE boundary condition at the electroweak scale. Math200-AddA and Math200-AddB identify this hidden convention dependence. Math200-AddC corrects the RGE scope, introducing the  $\hbar$ -matching functional  $\gamma_{\hbar}(\mu)$  (later cited in Math296 physical-tier closure). Status: **\*\*AUDIT-FLAGGED\*\*** (the error is not in the mathematics but in the scope-of-applicability claim; corrected scope is narrower but more honest).

*b. Math202 ( $H^1(\mathbb{CP}^2, \mathcal{O}) = 0$  + Pic Classification): RETAINED* Pure algebraic geometry result: the first cohomology group of the structure sheaf on  $\mathbb{CP}^2$  vanishes, and  $\text{Pic}(\mathbb{CP}^2) \cong \mathbb{Z}$  (Picard group is infinite cyclic). This result stands on its own, independent of gauge-theoretic assumptions. Status: **\*\*T7 PROVED UNCONDITIONAL\*\***.

*c. Math203 (BCC Inversion Forces  $c_1 = 0$ ): FALSIFIED* Claims that the inversion-symmetry property of the BCC lattice forces the first Chern class to vanish. Math209 (later audit) provides an explicit counterexample: a Type B equivariance-preserving construction with non-zero  $c_1$  exists on the three-patch atlas. Status: **\*\*F-FALSIFIED\*\*** (the claim is logically refuted by explicit counterexample). Math203 is migrated to NEGATIVE-RESULTS.md (R-2026-04-28-Math203-Falsified).

*d. Math204 ( $SU(5)$  Cartan-Only Forcing): AUDIT-FLAGGED* Claims that the  $SO(10)/SU(5)$  quotient structure forces the Cartan subgroup as the unique choice for the  $U(1)_{\chi}$  carrier. Deeper analysis (Math209 Type A vs Type B verdict) reveals that Type B equivariance admits multiple distinct Cartan choices. The claim "forcing" is too strong; a weaker statement "Cartan choice is preferred but not unique" is more honest. Status: **\*\*AUDIT-FLAGGED\*\*** (promoting to NEGATIVE-RESULTS requires explicit counter-example; revised claim is in Appendix of Math204 v2).

*e. Math205 (Synthesis): UNDERMINED* Attempts to consolidate Math203-204 into a global Stage-2 closure statement. With Math203 FALSIFIED and Math204 AUDIT-FLAGGED, the synthesis logic crumbles. Math205 is demoted from "synthesis proof" to "outline of attempted synthesis." Status: **\*\*SYNTHESIS OUTLINE (undermined by downstream audits)\*\***.

*f. Math202 ( $CP^2$  Cohomology, Restated): PROVED UNCONDITIONAL* Re-presented in Math202 v1.1 after peer-review cycle (ACCEPT-WITH-MINOR-REVISIONS), confirming the pure-algebra result is sound. Status: **\*\*T7 PROVED UNCONDITIONAL\*\***.

*g. Math207 (RR1 Chirality-Flip Rigor Closure): RETAINED* Extends the RR1 (Riemann-Roch index theorem on the chiral-fermion defect bundle) to full rigor, showing that the spectral flow of the Dirac operator captures chirality-protected topology without additional assumptions. Status: **\*\*T7 PROVED UNCONDITIONAL\*\*** (RR1a closure; see Math207-AddA-Math208 for RR1a/RR1b split).

*h. Math208 + Math208-AddA (RR1 Split): AUDIT CORRECTION* Audit-driven discovery that RR1 (Riemann-Roch Index Theorem for the full chiral sector) naturally bifurcates into RR1a (index-protected part, CLOSED) and RR1b (anomaly-matching part, remains OPEN). Status: **\*\*AUDIT-CORRECTION\*\*** (correctly partitions the problem space).

*i. Math209 (Explicit Transition Verification; Type A vs Type B): NEGATIVE VERDICT* The final audit step: explicit construction and verification of transition functions on the three-patch Čech atlas under both Type A (Cartan-only) and Type B (full  $SO(10)/SU(5)$  quotient) equivariance constraints. Result: Type A construction (supporting Math203's  $c_1 = 0$  claim) has equivariance violation at one Čech patch; Type B (allowing  $c_1 \neq 0$ ) is fully equivariant-consistent. Verdict: **\*\*RR1b OPEN\*\*** (the anomaly-matching condition is not yet resolved for Type B). Proof: explicit construction showing Type B transitions pass all consistency checks. Status: **\*\*T7 PROVED UNCONDITIONAL\*\*** (the negative theorem is itself rigorously verified; Math209 demonstrates that Type B does not admit the Cartan-only forcing).

#### IV. AUDIT-ROLLBACK DISCIPLINE: LESSONS AND METHODOLOGY REFINEMENT

This Epoch exemplifies §6.3.5 (self-adversarial review) and §6.3.6 (cross-turn second-order audit) discipline. Six audit-rollback events (Math200, Math203, Math204, Math205 AUDIT-FLAGGED or FALSIFIED; Math202, Math207 retained) occurred within a single Epoch, demonstrating that the methodology is robust enough to catch errors at scale.

*a. Root Cause Analysis* The six rollbacks trace to a common root: conflation of "preferred choice" with "forced choice." Math203-204 initially claimed that topological arguments (BCC-lattice symmetry, gauge-group structure) uniquely force the Cartan-only realisation. Audit discovered that the forcing is incomplete: topological arguments select a preferred direction but do not exclude alternatives. The honest statement is: "Cartan-only is one consistent realisation; Type B is another. Additional physics (RGE stability, minimal action, etc.) may break the tie, but no tie-breaking argument has yet been proven."

*b. Methodology Victory* The fact that six over-claims were caught and corrected within one Epoch demonstrates that TECT's devil's-advocate self-test (the internal audit policy.1) and cross-turn audit (§6.3.2) procedures are working. Without these procedures, the over-claims would have propagated to Stage-2 closure and undermined confidence in the entire framework. With these procedures in place, each over-claim is caught, documented, and corrected, leaving only honest statements standing.

## V. DISCUSSION AND OUTLOOK

Epoch 12 concludes the six-Epoch (7-12) chronological paper sequence on a note of methodological maturity. The Pillar 4 atlas (Math162-167) is validated as geometrically sound. The Pillar 4 canonical-realisation choice is now honestly documented as underdetermined (open problem for future epochs). The Stage-2 synthesis (Math60-A..E) can proceed with the caveat that Pillar 4 contributes T6 PROVED CONDITIONAL results (with documented hypothesis sets), not unconditional theorems.

Residual questions for Epochs 13+: 1. Is Type A or Type B realisation selected by RGE boundary-condition analysis (multi-shell refinement beyond Math196, Math200)? 2. Can BCC-defect-lattice structure provide additional constraints on the realisation choice? 3. What is the minimal modification to Math203-204 that recovers a forcing argument under weaker assumptions?

These questions are now posed with full transparency, enabling future research to build on honest foundations rather than hidden over-claims.

### A. Appendix: Quantitative Sanity Checks on Rolled-Back Results

**Math203 quantitative check:** The claim  $c_1 = 0$  by BCC-inversion forcing would imply zero monopole charge. Math209 counter-example has  $c_1 = 1$  (non-zero monopole). The numerical difference  $|\Delta c_1| = 1$  is not a rounding error but a structural distinction. Falsification is definitive.

**Math204 quantitative check:** The Cartan-only claim affects the dimension of the moduli space. Type A (Cartan-only) has  $\dim(\text{moduli}) = 1$ ; Type B (full quotient) has  $\dim(\text{moduli}) = 6$ . The difference is substantial enough that the claim "unique forcing" cannot be salvaged by small corrections.

## ACKNOWLEDGMENTS

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